Supplementary Material of "Accurate learning of long-range interatomic potentials by coupling Cartesian atomic cluster expansion and sum-of-Gaussians neural networks"

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I. DERIVATION OF EQS. (10)-(13) IN THE MAIN TEXT

We begin by discretizing Eq. (10) using the trapezoidal rule with uniformly spaced nodes $x_m = t_0 + m\zeta$, where $m \in \mathbb{Z}$, $\zeta > 0$ is the step size, and $t_0 \in \mathbb{R}$ being referred to as the starting point of trapezoidal rule:

$$\frac{1}{r^{2\alpha}} \approx \mathscr{F}_{\mathrm{SOG},\alpha}(r) = \zeta \sum_{m=-\infty}^{\infty} \mathscr{T}_{r,\alpha}(t_0 + m\zeta), \tag{I.1}$$

with $\mathscr{T}_{r,\alpha}(x) := (\Gamma(\alpha))^{-1} e^{-e^x r^2 + \alpha x}$. The bilateral series approximation (BSA) of $1/r$ is obtained by taking $t_0 = -\log(2s^2)$ and $\zeta = 2\log b$ with parameters $s > 0$ and $b > 1$. Let

$$\widehat{\mathscr{T}}_{r,\alpha}(k) = \int_{\mathbb{R}} \mathscr{T}_{r,\alpha}(x) e^{-\iota k x} dx = \frac{\Gamma(\alpha - \iota k)}{\Gamma(\alpha)} r^{2\iota k - 2\alpha} \tag{I.2}$$

be the Fourier transform of $\mathscr{T}_{r,\alpha}(x)$. By the Poisson summation formula, we have

$$\sum_{m\in\mathbb{Z}} \mathscr{T}_{r,\alpha}(t_0 + m\zeta) = \frac{1}{\zeta} \sum_{m\in\mathbb{Z}} \widehat{\mathscr{T}}_{r,\alpha}(2\pi m/\zeta) e^{\iota 2\pi t_0 m\zeta}. \tag{I.3}$$

Since $\widehat{\mathscr{T}}_{r,\alpha}(0) = \int_{\mathbb{R}} \mathscr{T}_{r,\alpha}(x)dx = 1/r^{2\alpha}$, we have

$$\begin{aligned} \left|\frac{1}{r^{2\alpha}} - \mathscr{F}_{\mathrm{SOG},\alpha}\right| &= \left|\widehat{\mathscr{T}}_{r,\alpha}(0) - \sum_{m\in\mathbb{Z}} \widehat{\mathscr{T}}_{r,\alpha}\left(\frac{2\pi m}{\zeta}\right) e^{\iota 2\pi t_0 m\zeta}\right| \\ &\le \sum_{m\neq 0} \left|\widehat{\mathscr{T}}_{r,\alpha}\left(\frac{2\pi m}{\zeta}\right)\right| \\ &\le \frac{1}{r^{2\alpha}} \sum_{m\neq 0} \left|\frac{\Gamma(\alpha - 2\pi\iota m/\zeta)}{\Gamma(\alpha)}\right|, \end{aligned} \tag{I.4}$$

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where we used Eq. (I.2) in the last step. The sum is dominated by the $m = 1$ and $m = -1$ terms, which have equal magnitude. Using the asymptotic decay of the Gamma function as $\zeta \to 0$,

$$|\Gamma(\alpha - 2\pi\iota/\zeta)| \sim (2\pi)^{1/2} \left(\alpha^2 + \frac{(2\pi)^2}{\zeta^2}\right)^{\frac{2\alpha-1}{4}} e^{-\frac{\pi^2}{\zeta}}, \tag{I.5}$$

which follows from Stirling's formula (see Appendix B of the arXiv version of Ref.[1] for a derivation), we combine Eqs. (I.4)-(I.5) to obtain the relative-error bound as $\zeta \to 0$:

$$\frac{\left|1/r^{2\alpha} - \mathscr{F}_{\mathrm{SOG},\alpha}(r)\right|}{|1/r^{2\alpha}|} \leq \frac{(2\pi)^{1/2}}{\Gamma(\alpha)} \left(\alpha^2 + \frac{(2\pi)^2}{\zeta^2}\right)^{\frac{2\alpha-1}{4}} e^{-\frac{\pi^2}{\zeta}}. \tag{I.6}$$

Finally, setting $\alpha = 1/2$ and $\zeta = 2\log b$ in Eq. (I.6) obtains the error bound for the $1/r$ BSA series reported in Eq. (12) in the main text (the subscript α of $\mathscr{F}_{\mathrm{SOG},\alpha}$ is omitted in Eq. (12)).

Next, we discuss how to choose the BSA parameters b and s. Following the original BSA paper[1], the $1/r$ kernel can be decomposed into the sum of a short-range part and a long-range part:

$$\frac{1}{r} = \mathscr{N}_{\mathrm{SOG}}(r) + \mathscr{F}^M_{\mathrm{SOG}}(r), \tag{I.7}$$

where

$$\mathscr{N}_{\mathrm{SOG}}(r) = \begin{cases} \dfrac{1}{r} - \mathscr{F}^M_{\mathrm{SOG}}(r), & \text{if } r < r_{\mathrm{cut}} \\ \\ 0, & \text{if } r \geq r_{\mathrm{cut}} \end{cases} \tag{I.8}$$

and the long-range part is the BSA series truncated to $0 \leq m \leq M-1$,

$$\mathscr{F}^M_{\mathrm{SOG}}(r) = \frac{2\log b}{\sqrt{2\pi s^2}} \sum_{m=0}^{M-1} b^{-m} e^{-\frac{r^2}{2b^{2m}s^2}}. \tag{I.9}$$

Here, r_{cut} is a cutoff radius for the short-range part. By Eq. (12) in the main text, the asymptotic relative error bound depends only on the parameter b. Given a target tolerance ε, we choose b such that

$$2\sqrt{2}e^{-\frac{\pi^2}{2\log b}} \approx \varepsilon. \tag{I.10}$$

We then choose s to enforce C^0-continuity at r_{cut}, i.e.,

$$\frac{1}{r_{\mathrm{cut}}} - \mathscr{F}^M_{\mathrm{SOG}}(r_{\mathrm{cut}}) = 0, \tag{I.11}$$

To solve Eq. (I.11), we use the following property:

$$\mathscr{F}^M_{\mathrm{SOG}}(r_{\mathrm{cut}}) = \frac{1}{s}\mathscr{F}^{M,1}_{\mathrm{SOG}}\left(\frac{r_{\mathrm{cut}}}{s}\right), \tag{I.12}$$

where

$$\mathscr{F}^{M,1}_{\mathrm{SOG}}(r) := \frac{2\log b}{\sqrt{2\pi}} \sum_{m=0}^{M-1} b^{-m} e^{-\frac{r^2}{2b^{2m}}} \tag{I.13}$$

is defined by setting $s \equiv 1$ in Eq. (I.9). Let $r_0 = r_{\mathrm{cut}}/s$. Then Eq. (I.11) is equivalent to

$$r_0 \mathscr{F}_{\mathrm{SOG}}^{M,1}(r_0) - 1 = 0. \tag{I.14}$$

Note that the solution for r_0 is not unique. We take the smallest $r_0 > 0$ as suggested in Ref.[1]. For $b = 2$, this gives $\varepsilon \approx 10^{-3}$ and $r_0 = 1.9892536839080267$, which we find sufficient for our SOG-Net experiments. For other choices of b and ε, we refer the reader to Ref.[1] for further details.

We explain the rationale for using Eq. (13) in the main text to parametrize the sum-of-Gaussians (SOG) multiplier. Consider a periodic system of N atoms located at $\boldsymbol{r}_1, \cdots, \boldsymbol{r}_N$ in a domain Ω with side lengths L_x, L_y, and L_z. By the BSA-based decomposition of the $1/r$ kernel, the Coulomb energy can be written as $E_{\mathrm{coul}} := E_{\mathscr{N}} + E_{\mathscr{F}} - E_{\mathrm{self}}$, where

$$E_{\mathscr{N}} = \frac{1}{2} \sum_{\boldsymbol{n} \in \mathbb{Z}^3} {}' \sum_{i,j} q_i q_j \mathscr{N}_{\mathrm{SOG}}(|\boldsymbol{r}_{ij} + \boldsymbol{n} \circ \boldsymbol{L}|), \tag{I.15}$$

$$E_{\mathscr{F}} = \frac{1}{2} \sum_{\boldsymbol{n} \in \mathbb{Z}^3} \sum_{i,j} q_i q_j \mathscr{F}_{\mathrm{SOG}}^{M}(|\boldsymbol{r}_{ij} + \boldsymbol{n} \circ \boldsymbol{L}|), \tag{I.16}$$

and

$$E_{\mathrm{self}} = \frac{1}{2} \sum_{\substack{i,j \\ i=j}} q_i q_j \mathscr{F}_{\mathrm{SOG}}^{M}(|\boldsymbol{r}_{ij}|) = \frac{\log b}{\sqrt{2\pi s^2}} \left(\sum_{m=0}^{M-1} b^{-m} \right) \sum_{i=1}^{N} q_i^2 = \frac{(b - b^{-(M-1)}) \log b}{\sqrt{2\pi} s (b-1)} \sum_{i=1}^{N} q_i^2 \tag{I.17}$$

are short-range, long-range, and self-energy parts, respectively. Here, $\boldsymbol{r}_{ij} := \boldsymbol{r}_i - \boldsymbol{r}_j$, the prime in Eq. (I.15) excludes the term $i = j$ and $\boldsymbol{n} = \boldsymbol{0}$, "$\circ$" denotes the Hadamard product of two vectors, and q_i is the charge associated with atom i. Applying the Poisson summation formula to Eq. (I.16), the long-range part of Coulomb energy can be written in Fourier space as:

$$E_{\mathscr{F}} = \frac{1}{2V} \sum_{\boldsymbol{k}} \widehat{\mathscr{F}}_{\mathrm{SOG}}^{M}(\boldsymbol{k}) \left| S(\boldsymbol{k}) \right|^2, \tag{I.18}$$

where $\boldsymbol{k} = 2\pi(m_x/L_x, m_y/L_y, m_z/L_z)$ denotes the Fourier modes and runs over all integer triples (m_x, m_y, m_z),

$$\widehat{\mathscr{F}}_{\mathrm{SOG}}^{M}(\boldsymbol{k}) = \int_{\mathbb{R}^3} \mathscr{F}_{\mathrm{SOG}}^{M}(|\boldsymbol{r}|) e^{-\iota \boldsymbol{k} \cdot \boldsymbol{r}} d\boldsymbol{r} = 4\pi s^2 \log b \sum_{m=0}^{M-1} b^{2m} e^{-b^{2m} s^2 k^2/2} \tag{I.19}$$

is the three-dimensional Fourier transform of $\mathscr{F}_{\mathrm{SOG}}^{M}$, and

$$S(\boldsymbol{k}) := \sum_{i=1}^{N} q_i e^{\iota \boldsymbol{k} \cdot \boldsymbol{r}_i} \tag{I.20}$$

denotes the charge structure factor. From Eq. (I.19), the decay of $\widehat{\mathscr{F}}_{\mathrm{SOG}}^{M}$ is controlled by b. Taking $b \to 1$ improves the BSA approximation error, but it also requires a finer k-space resolution (more Fourier modes) to evaluate Eq. (I.18) accurately. In practice, we find $b = 2$ provides a good trade-off between training accuracy and computational cost.

A detailed derivation of Eqs. (I.15)-(I.20) is provided in Ref.[2]. Comparing Eq. (I.18) with Eq. (6) in the main text shows that initializing the learned Fourier multiplier $\widehat{f}_{\boldsymbol{\theta}_{\mathrm{SOG}}}$ by the BSA form $\widehat{\mathscr{F}}_{\mathrm{SOG}}^{M}$ amounts to starting from the perfect $1/r$ Coulomb long-range energy. This physics-guided initialization acts as a simple normalization prior: it places the initial Fourier multiplier close to the expected long-range decay, thus speeding up training and helping prevent overfitting.

II. MODEL REDUCTION FOR AN SOG APPROXIMATION

Suppose we have an M-term sum-of-Gaussians series

$$f(k) = \sum_{m=0}^{M-1} w_m e^{-k^2/s_m^2}, \qquad k \geq 0. \tag{II.1}$$

Model reduction aims to construct a lower-order SOG with $T < M$ terms that approximates the function f in Eq. (II.1) within a prescribed tolerance, i.e.,

$$\max_{k\geq 0} \left| \sum_{m=0}^{M-1} w_m e^{-k^2/s_m^2} - \sum_{\ell=0}^{T-1} c_\ell e^{-k^2/\eta_\ell^2} \right| \leq \varepsilon. \tag{II.2}$$

Many reduction techniques are available for SOG expansions[3]. Here we outline a balanced-truncation approach[4], which is widely used in practice.

We first apply the change of variables $x = k^2$ so that the SOG becomes a sum of exponentials in x, and then take the Laplace transform:

$$\mathscr{L}\left[\sum_{m=0}^{M-1} w_m e^{-x/s_m^2}\right] = \sum_{m=0}^{M-1} \frac{w_m}{z + 1/s_m^2}. \tag{II.3}$$

This sum-of-poles representation can be written as the transfer function of a linear dynamical system,

$$\sum_{m=0}^{M-1} \frac{w_m}{z + 1/s_m^2} = \boldsymbol{C}(z\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}, \tag{II.4}$$

where

$$\boldsymbol{A} = -\mathrm{diag}\left(1/s_0^2, 1/s_1^2, \cdots, 1/s_{M-1}^2\right) \tag{II.5}$$

is a $M \times M$ diagonal matrix, and

$$\boldsymbol{B} = \left(\sqrt{|w_0|}, \sqrt{|w_1|}, \cdots, \sqrt{|w_{M-1}|}\right)^* \qquad \text{and} \qquad \boldsymbol{C} = \left(w_0/\sqrt{|w_0|}, \cdots, w_{M-1}/\sqrt{|w_{M-1}|}\right) \tag{II.6}$$

are a column and a row vector, respectively. Here "$*$" denotes transpose.

Next, we solve the Lyapunov equations,

$$\boldsymbol{AP} + \boldsymbol{PA}^* = -\boldsymbol{BB}^* \quad \text{and} \quad \boldsymbol{AQ} + \boldsymbol{QA}^* = -\boldsymbol{C}^*\boldsymbol{C}, \tag{II.7}$$

and compute Cholesky factors $\boldsymbol{P} = \boldsymbol{S}\boldsymbol{S}^*$ and $\boldsymbol{Q} = \boldsymbol{L}\boldsymbol{L}^*$. We then compute the singular value decomposition

$$\boldsymbol{S}^*\boldsymbol{L} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*, \qquad \boldsymbol{\Sigma} = \mathrm{diag}(\sigma_0, \sigma_1, \ldots, \sigma_{M-1}), \tag{II.8}$$

where σ_i are the Hankel singular values. We form the balancing transformation matrix $\boldsymbol{X} = \boldsymbol{S}\boldsymbol{U}\boldsymbol{\Sigma}^{-\frac{1}{2}}$ and define

$$\widetilde{\boldsymbol{A}} = \boldsymbol{X}^{-1}\boldsymbol{A}\boldsymbol{X}, \qquad \widetilde{\boldsymbol{B}} = \boldsymbol{X}^{-1}\boldsymbol{B}, \qquad \widetilde{\boldsymbol{C}} = \boldsymbol{C}\boldsymbol{X}. \tag{II.9}$$

The reduced T-order model is obtained by taking the leading $T \times T$, $T \times 1$, and $1 \times T$ blocks of $\widetilde{\boldsymbol{A}}$, $\widetilde{\boldsymbol{B}}$, and $\widetilde{\boldsymbol{C}}$, denoted by $\boldsymbol{A}'$, $\boldsymbol{B}'$, and $\boldsymbol{C}'$. We choose T such that the standard balanced-truncation error bound satisfies

$$\sup_{z=\iota\mathbb{R}} \left|\boldsymbol{C}'(z\boldsymbol{I} - \boldsymbol{A}')^{-1}\boldsymbol{B}' - \boldsymbol{C}(z\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}\right| \le 2\sum_{j=T}^{M-1} \sigma_j \le \varepsilon. \tag{II.10}$$

Finally, we compute an eigen-decomposition $\boldsymbol{A}' = \boldsymbol{X}'\boldsymbol{\Lambda}\boldsymbol{X}'^{-1}$. Let $\eta_\ell = 1/\sqrt{\Lambda_{\ell\ell}}$ for $\ell = 0, 1, \ldots, T-1$. Define $\boldsymbol{B}'' = \boldsymbol{X}'^{-1}\boldsymbol{B}'$ and $\boldsymbol{C}'' = \boldsymbol{C}'\boldsymbol{X}'$, and set $c_\ell = B''_\ell C''_\ell$. This yields the reduced SOG in Eq. (II.2). Similar constructions for SOG-based kernels have been widely studied; see, e.g., Refs.[5–8]. An example implementation is available at https://github.com/DuktigYajie/SOG-Net/blob/main/CACE-SOG/model_reduction.

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